

Linear algebra – 5

Bilinear and quadratic forms

Homework

12 сентября 2026 года

Theoretical minimum. Congruence and the law of inertia; Sylvester's criterion; the Gram matrix and its determinant as a squared volume; simultaneous diagonalisation of a pair of forms; the Schur complement; Hadamard's and Minkowski's determinant inequalities.

1. (Hadamard) Prove that $|\det A| \leq \prod_{j=1}^n \|A_j\|$ for the columns A_j of a real matrix A , with equality exactly when the columns are pairwise orthogonal or one of them is zero. Deduce that $\det A \leq \prod a_{ii}$ for positive semidefinite A .
2. (Romania national olympiad 1998) Let $n \geq 2$ and let $A_0, A_1, \dots, A_n \in M_2(\mathbb{R})$ be nonzero matrices with $A_0 \neq aE$ for every $a \in \mathbb{R}$ and $A_0 A_k = A_k A_0$ for $k = 1, \dots, n$.
 - a) Prove that $\det(\sum_{k=1}^n A_k^2) \geq 0$.
 - b) Prove that if this determinant vanishes and $A_2 \neq aA_1$ for every $a \in \mathbb{R}$, then $\sum_{k=1}^n A_k^2 = O_2$.
3. (VJIMC 2007, cat. II) Let A be a real $n \times n$ matrix with $A + A^T = E$. Show that $\det A > 0$.
4. (SEEMOUS 2013) Let $A \in M_n(\mathbb{C})$ and $a \in \mathbb{C}$, $a \neq 0$, be such that $A - A^* = 2aE$, where $A^* = \overline{A}^T$.
 - a) Show that $|\det A| \geq |a|^n$.
 - b) Show that equality forces $A = aE$.

5. (SEEMOUS 2019) Let m, n be positive integers. Prove that for any $A_1, \dots, A_m \in M_n(\mathbb{R})$ there exist $\varepsilon_1, \dots, \varepsilon_m \in \{-1, 1\}$ with

$$\operatorname{tr}((\varepsilon_1 A_1 + \dots + \varepsilon_m A_m)^2) \geq \operatorname{tr}(A_1^2) + \dots + \operatorname{tr}(A_m^2).$$

6. (SEEMOUS 2010) Prove that

- a) for every $A \in M_2(\mathbb{R})$ there are $B, C \in M_2(\mathbb{R})$ with $A = B^2 + C^2$;
- b) there are no $B, C \in M_2(\mathbb{R})$ with $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = B^2 + C^2$ and $BC = CB$.

7. (Schur complement) Let A be positive definite. Prove that $\begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \succeq 0$ if and only if $C - B^T A^{-1} B \succeq 0$, and deduce the Fischer inequality $\det \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \leq \det A \cdot \det C$.

8. (Ky Fan) Prove that $\det(tA + (1 - t)B) \geq (\det A)^t (\det B)^{1-t}$ for positive definite A, B and $t \in [0, 1]$; i.e. $\log \det$ is concave on the positive definite cone.

9. (Minkowski) Prove that $\det(A + B)^{1/n} \geq (\det A)^{1/n} + (\det B)^{1/n}$ for positive semidefinite A, B , and explain in what sense this is the matrix form of the Brunn–Minkowski inequality.

10. (IMC) Let $v_0 = 0$ and let $v_1, \dots, v_{n+1} \in \mathbb{R}^n$ be such that the Euclidean distance $|v_i - v_j|$ is rational for all $0 \leq i, j \leq n + 1$. Prove that $v_1, \dots, v_{n+1}$ are linearly dependent over $\mathbb{Q}$.